

WEEK 1 · SESSION 2

Differential Drive

How the XRP actually moves

Two wheels. Two motor commands. Every path the robot can make.

What is differential drive?

INDEPENDENT

Left and right wheels each have their own motor.

NO STEERING LINKAGE

The wheels do not pivot like a car's front wheels.

DIFFERENCE CREATES MOTION

Speed and direction on each side determine the path.



equal → straight



different → curve



opposite → spin

Three motion patterns

STRAIGHT

$$v_L = v_R$$

Same speed and same direction.

CURVE

$$v_L \neq v_R$$

The robot curves toward the slower wheel.

SPIN IN PLACE

$$v_L = -v_R$$

Equal magnitudes, opposite directions.

A larger speed difference produces a tighter turn.

Motion → motor commands

Straight

(0.5, 0.5)

Curve left

(0.2, 0.5)

Spin

(0.5, -0.5)

```
from XRPLib.differential_drive import DifferentialDrive
drive = DifferentialDrive.get_default_differential_drive()

drive.set_effort(0.5, 0.5)
drive.set_effort(0.2, 0.5)
drive.set_effort(0.5, -0.5)
drive.stop()
```

Effort: -1.0 full reverse · 0 stopped · +1.0 full forward

What the two wheel speeds mean

$$v = \frac{v_R + v_L}{2} \quad \omega_{\text{robot}} = \frac{v_R - v_L}{b}$$

FORWARD SPEED

The wheel-speed average moves the robot's center.

ROBOT YAW RATE

The wheel-speed difference rotates it; **b** is wheel spacing.

Sum controls translation. Difference controls rotation.

One turn center, three circles

SHARED CENTER

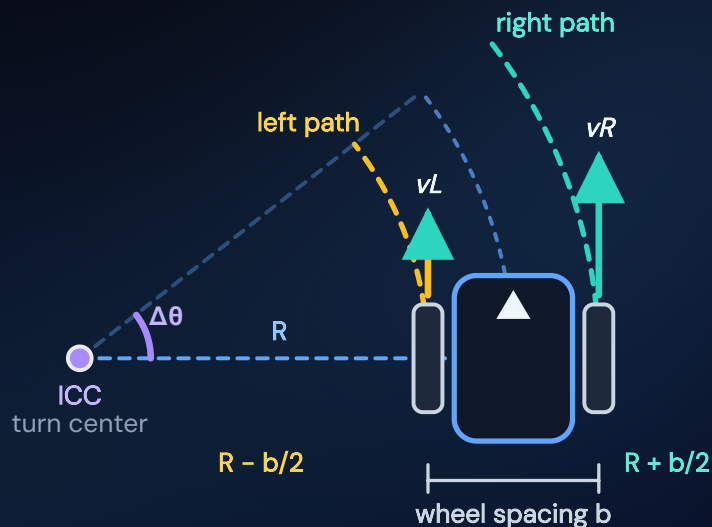
With constant wheel speeds, the robot turns about an **instantaneous center of curvature (ICC)**.

INNER WHEEL

The left wheel traces the smaller circle: $R - b/2$.

OUTER WHEEL

The faster right wheel traces the larger circle: $R + b/2$.



Geometry adapted from [Han, Choi & Lee \(2008\)](#).

Same angle, different distances

$$\Delta s_L = \left(R - \frac{b}{2}\right) \Delta\theta$$
$$\Delta s_R = \left(R + \frac{b}{2}\right) \Delta\theta$$

Both wheels sweep the same angle $\Delta\theta$ in the same time Δt .

LEFT WHEEL

$$v_L = \left(R - \frac{b}{2}\right) \omega_{\text{robot}}$$

RIGHT WHEEL

$$v_R = \left(R + \frac{b}{2}\right) \omega_{\text{robot}}$$

Here, $\omega_{\text{robot}} = \Delta\theta / \Delta t$ is the robot's yaw rate—not a wheel's angular speed.

Solve for the turn radius

ADD THE WHEEL EQUATIONS

$$v_R + v_L = 2R\omega_{\text{robot}}$$

SUBTRACT THE WHEEL EQUATIONS

$$v_R - v_L = b\omega_{\text{robot}}$$

DIVIDE, THEN SOLVE FOR R

$$R = \frac{b(v_R + v_L)}{2(v_R - v_L)}$$

Use $|R|$ for the circle's radius; the sign tells the turn direction.

SAME SPEEDS

$$v_R = v_L$$

$R \rightarrow \infty$ · straight

OPPOSITE SPEEDS

$$v_R = -v_L$$

$R = 0$ · spin

EXAMPLE

$$b = 0.16 \text{ m}$$

$$v_L = 0.20, v_R = 0.40$$

$$R = 0.24 \text{ m}$$

Two numbers control every path

Match

go straight

Offset

follow a curve

Oppose

spin in place